REAL-TIME EMBEDDED SYSTEM
EXPERIMENT NO. 4

APPLIANCE CONTROLLER

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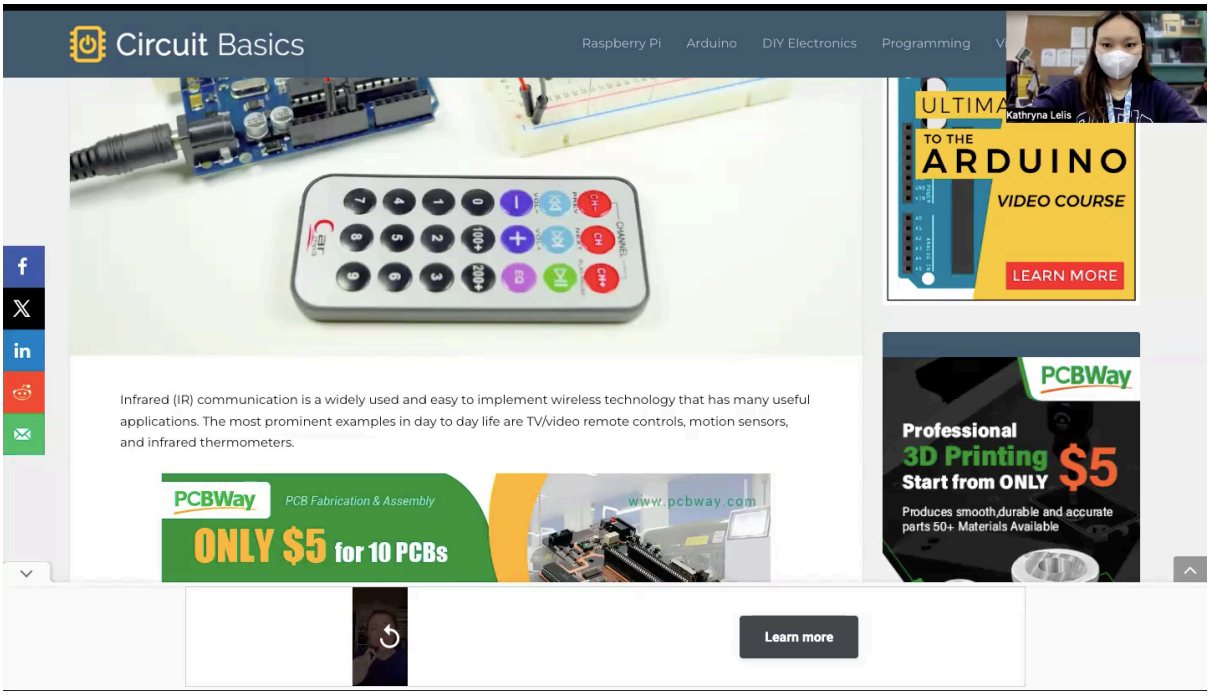
DISCUSSION

Experiment 4 uses a lot of component that I usually dont work with such as the stepper motor and IR remote. therefore, the first step in creating a functional appliance controller was to research how each component works and can be integrated with one another.



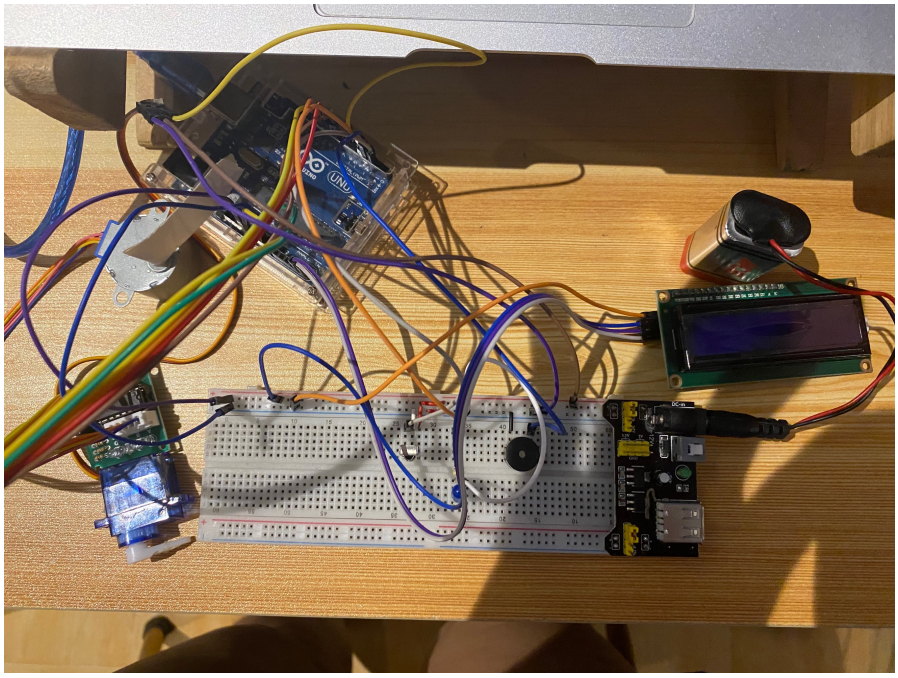
After researching, I discovered that I needed to check the remote buttons' HEX values to integrate

them with other components. Using code available online, I tested the remote and recorded the HEX values of my controller. I then used these values to successfully implement an LED on-and-off function.

```
/* Key   Code
 CH-   BA45FF00
 CH    B946FF00
 CH+   B847FF00
 <<    BB44FF00
 >>    BF40FF00
 >||   BC43FF00
 -     F807FF00
 +     EA15FF00
 EQ    F609FF00
 100+  E619FF00
 200+  F20DFF00
 0     E916FF00
 1     F30CFF00
 2     E718FF00
 3     A15EFF00
 4     F708FF00
 5     E31CFF00
 6     A55AFF00
 7     BD42FF00
 8     AD52FF00
 9     B54AFF00 */
```

Having the led test function to be successful, i continued with the buzzer that I dont have any difficulty working with. However, when it came time to work with the motors, I had to research and

understand their library codes. Although the code itself showed no errors, I encountered problems when the servo and stepper motors were connected to the Arduino. The issue revolved around current and voltage: the motors' rotations were not as smooth when operating together as they were individually. Adding the LCD further trouble the voltage and current issues. Nonetheless, they work as I added a power module, and a battery yet still lags from time to time.



Completing all the code of the component, i continue to test and debug problems or things I dont like in the system. Then ensure if all objectives are achieved. Once ensured its all done.

CONCLUSION

This experiment has successfully shown how an appliance controller works in real time with the integration of several components like an IR remote, steper motor, servo motor, LCD, and buzzer. After rigorous research, testing, and debugging, I have deployed and maintained the behavior of every single component. Current and voltage problems, especially in a situation where several components were working together, were overcome by employing an external power module and an external battery.

This experiment provided an opportunity to apply some of the concepts learned in the areas of logic circuits, switching theory, and

microprocessor programming. Using the Arduino kit, I have actually developed a real-time embedded system demonstrating in practical uses of theoretical principles.

The objectives of the experiment, however, were achieved with the system working as it is supposed to work.